

An Analysis of Random forest

Random forest

Random forest -- Part 1

Prefiltering: Eliminate features that are mostly just noise.
Enriched Random Forest (ERF): Use weighted random sampling instead of simple random sampling at each node of each tree, giving greater weight to features that appear to be more informative. Tree-weighted random forest (TWRF): Give more weight to more accurate trees.

Random forest -- Part 2

Mean decrease in impurity feature importance This approach to feature importance for random forests considers as important the variables which decrease a lot the impurity during splitting. It is described in the book Classification and Regression Trees by Leo Breiman and is the default implementation in sci-kit learn and R. The definition is:
$$\text{unnormalized average importance}(x) = \frac{1}{n_T} \sum_{i=1}^n \sum_{\text{node } j \in T_i} \text{split variable}(j) \times p_{T_i}(j) \Delta i_{T_i}(j)$$
 , where $p_{T_i}(j)$ is the proportion of samples in T_i that are split by variable j , and $\Delta i_{T_i}(j)$ is the decrease in impurity for node j in tree T_i .

Random forest -- Part 3

Preliminaries: Centered forests Centered forest is a simplified model for Breiman's original random forest, which uniformly selects an attribute among all attributes and performs splits at the center of the cell along the pre-chosen attribute. The algorithm stops when a fully binary tree of level k is built, where $k \in \mathbb{N}$ is a parameter of the algorithm.

Random forest -- Part 4

Variants Instead of decision trees, linear models have been proposed and evaluated as base estimators in random forests, in particular multinomial logistic regression and naive Bayes classifiers. In cases that the relationship between the predictors and the target variable is linear, the base learners may have an equally high accuracy as the ensemble learner.

Random forest -- Part 5

$$\sigma = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (f_b(x') - \bar{f})^2}.$$

Random forest -- Part 6

Centered KeRF The construction of Centered KeRF of level k is the same as for centered forest, except that

predictions are made by $\mathbf{m} \sim \mathbf{M}$, $\mathbf{n}(\mathbf{x}, \Theta_1, \dots, \Theta_M)$ $\{\tilde{\mathbf{m}}\}_{M,n}(\mathbf{x}, \Theta_1, \dots, \Theta_M)\}$, the corresponding kernel function, or connection function is

Random forest -- Part 7

Disadvantages While random forests often achieve higher accuracy than a single decision tree, they sacrifice the intrinsic interpretability of decision trees. Decision trees are among a fairly small family of machine learning models that are easily interpretable along with linear models, rule-based models, and attention-based models. This interpretability is one of the main advantages of decision trees. It allows developers to confirm that the model has learned realistic information from the data and allows end-users to have trust and confidence in the decisions made by the model. For example, following the path that a decision tree takes to make its decision is quite trivial, but following the paths of tens or hundreds of trees is much harder. To achieve both performance and interpretability, some model compression techniques allow transforming a random forest into a minimal "born-again" decision tree that faithfully reproduces the same decision function. Another limitation of random forests is that if features are linearly correlated with the target, random forest may not enhance the accuracy of the base learner. Likewise in problems with multiple categorical variables.

Random forest -- Part 8

$$K_{\{k\}^{\{uf\}}(\mathbf{0}, \mathbf{x})} = \sum_{k_1, \dots, k_d} \sum_{j=1}^d k_j = k_1! k_1! \dots k_d! (1-d)^k \prod_{m=1}^d (1 - |x_m|)^{\sum_{j=0}^k m - 1} (-\ln |x_m|)^j j! \text{ for all } x \in [0, 1]^d.$$

Random forest -- Part 9

$$|m_{\mathbf{M},n}(\mathbf{x}) - m \sim \mathbf{M},n(\mathbf{x})| \leq b_{\mathbf{n}} - a_{\mathbf{n}} \text{ and } m \sim \mathbf{M},n(\mathbf{x}).$$

Random forest -- Part 10

ExtraTrees Adding one further step of randomization yields extremely randomized trees, or ExtraTrees. As with ordinary random forests, they are an ensemble of individual trees, but

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there are two main differences: (1) each tree is trained using the whole learning sample (rather than a bootstrap sample), and (2) the top-down splitting is randomized: for each feature under consideration, a number of random cut-points are selected, instead of computing the locally optimal cut-point (based on, e.g., information gain or the Gini impurity). The values are chosen from a uniform distribution within the feature's empirical range (in the tree's training set). Then, of all the randomly chosen splits, the split that yields the highest score is chosen to split the node. Similar to ordinary random forests, the number of randomly selected features to be considered at each node can be specified. Default values for this parameter are $\sqrt{p}$ for classification and p for regression, where p is the number of features in the model.

Random forest -- Part 11

$$P \left[\left| \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{x}_i, \Theta) - \mathbb{E} \ell(\mathbf{x}, \Theta) \right| \geq \varepsilon \right] \leq \frac{1}{2} e^{-2n\varepsilon^2}$$

Random forest -- Part 12

In k -NN, $W(\mathbf{x}_i, \mathbf{x}') = \frac{1}{k}$ if $\mathbf{x}_i$ is one of the k points closest to $\mathbf{x}'$, and zero otherwise. In a tree, $W(\mathbf{x}_i, \mathbf{x}') = \frac{1}{k'}$ if $\mathbf{x}_i$ is one of the k' points in the same leaf as $\mathbf{x}'$, and zero otherwise. Since a forest averages the predictions of a set of m trees with individual weight functions W_j , its predictions are $\hat{y} = \frac{1}{m} \sum_{j=1}^m \sum_{i=1}^n W_j(\mathbf{x}_i, \mathbf{x}') y_i = \sum_{i=1}^n \left(\frac{1}{m} \sum_{j=1}^m W_j(\mathbf{x}_i, \mathbf{x}') \right) y_i$.